SJ INNOVATION · ASSESSMENT 3 (CONTINUATION OF ASSESSMENT 2)

QuickBite — Level 3: integrated, observable, production-grade AI

Food Ordering & Delivery · real integrations, multi-agent AI, and honest evaluation

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Due	**Friday, 9 October 2026** — submit on **Dev Control Tower**
Your app	https://quickbite-nine-phi.vercel.app/

Where you got to in Assessment 2

A RIO tool-calling edge agent (cart/checkout/track/escalate), smart rider-assignment + kitchen-load automations with an audit trail, Ollama embeddings, and a 97.4% menu-category classifier. Your honest RAG eval showed weak retrieval (37.5% hit-rate).

> **Assessment 3 in one line:** take the same app and make it **integrated** (real third-party services + a campaign/CRM module), **intelligent** (multi-agent with guardrails), **measured** (a real agent/RAG evaluation harness, hybrid + re-rank), and **production-grade** (CI/CD, observability, security).

> You already use Ollama — go for the bonus: **fine-tune or locally serve** a model the agent actually uses, and get the live integration working end-to-end.

Your seven tracks

1 Third-party integration (free tier) — 20 pts

Connect **≥2 real external services** relevant to your app, with **one outbound action** and **one inbound webhook** (verify the signature):

> **Twilio** (SMS/WhatsApp order updates), **Google Maps** (rider routing/ETA), and a **payment webhook** — one outbound + one verified inbound.

2 Campaign / CRM module — 15 pts

> A **loyalty / offers campaign** module: customer segments (new / frequent / churned), an offer-campaign builder, and redemption tracking.

Either sync to HubSpot or build it custom — it must have segments, a campaign/outreach action, and tracking.

3 Multi-agent orchestration + guardrails — 20 pts

> Grow RIO into a **multi-agent** system under a supervisor: RIO (ordering) + a **dispatch** agent (rider assignment/ETA) + a **support** agent (complaints/refunds), with guardrails and memory.

Required: a supervisor/router over specialist agents, shared **memory**, and **guardrails** (validation, PII redaction, low-confidence refusal, structured outputs).

4 Agent & RAG evaluation harness — 15 pts

Priority track — fix your RAG. Move to **hybrid search (vector + full-text) with re-ranking** and better chunking, scale to **≥20k records**, and re-run your eval: target a clear jump over 37.5% (aim ≥80% hit-rate on matching). Report before/after honestly.

Deliver numbers: agent task-success %, and RAG hit-rate / precision@k + faithfulness / answer-relevance, **side-by-side with your Assessment-2 results**.

5 Model improvement / fine-tune — 10 pts

> Predict **order-cancellation risk** with precision/recall/**F1** (target ≥0.80).

Report precision / recall / F1 (not just accuracy). Fine-tuning (LoRA / Ollama Modelfile) earns bonus.

6 CI/CD, observability & security — 15 pts

GitHub Actions running your full test suite on push; **error tracking + logs/metrics** (e.g. Sentry free); and a **Supabase advisor pass** fixing RLS/security findings, plus rate-limiting / caching.

7 Architecture docs, demo & Dev CT submission — 5 pts

An architecture diagram, the evaluation report, a short demo, and submission on **Dev Control Tower** with an `assessment-3` tag.

Tooling & how-to (all free tiers)

Third-party services (pick what fits your app)

- **HubSpot** free CRM — contacts, deals, lists; REST API + private-app token.
- **Email:** Resend or SendGrid free tier. **SMS/WhatsApp:** Twilio trial. **Calendar:** Google Calendar API. **Payments:** Stripe test mode + webhooks. **Chat:** Slack incoming webhooks.
- Build **one outbound call** (you call them) and **one inbound webhook** (they call you — verify the signature).

Deeper AI

- **Multi-agent:** a supervisor/router that delegates to specialist agents (each with its own tools), or a planner→executor loop. Add **memory** and **guardrails** (input/output validation, PII redaction, refuse/clarify on low confidence, structured JSON outputs).
- **Hybrid RAG:** combine vector + keyword (BM25 / Postgres full-text) retrieval, then **re-rank** the merged candidates (cross-encoder or an LLM re-ranker). Add **citations**.
- **Evaluation:** a labelled scenario set. For the agent report **task-success rate**; for RAG report **hit-rate / precision@k** plus **faithfulness** (is the answer grounded?) and **answer-relevance**. Always compare against your Assessment-2 baseline.
- **Model improvement:** report **precision / recall / F1** (not just accuracy); optionally **fine-tune** a small model (LoRA on free Colab, or a custom Ollama `Modelfile`) and compare to the baseline.

Production engineering

- **CI/CD:** GitHub Actions running the full test suite on every push/PR.
- **Observability:** error tracking (Sentry free) + structured logs/metrics; a small internal metrics view.
- **Security:** run Supabase advisors, fix RLS/security findings, add rate-limiting & caching.

How you will be scored (Assessment 3)

Full point breakdown is in the shared **Assessment 3 Scoring Parameters** sheet. Build against each parameter:

Parameter	Max
Third-party integration (≥2 services, 1 outbound + 1 inbound webhook)	20
Campaign / CRM module (segments, campaigns, tracking)	15
Multi-agent orchestration + guardrails + memory	20
Agent & RAG evaluation harness (hybrid + re-rank, metrics vs A2 baseline)	15
Model improvement (F1 reported, target; or fine-tune vs baseline)	10
CI/CD + observability + security (RLS advisor) pass	15
Architecture docs, demo & Dev Control Tower submission	5
Total	**100**
Bonus — live integration working in production; fine-tuned / local model used by the agent	+10

What to submit on Dev Control Tower

1. Updated **live URL** + **GitHub** (tag an `assessment-3` release this time).
2. **Architecture diagram** showing agents, integrations and data flow.
3. **Evaluation report** — agent task-success + RAG metrics, **side-by-side vs your Assessment-2 numbers**.
4. **Model metrics** (precision/recall/F1; fine-tune comparison if attempted).
5. **CI run link** (green) + a note on what observability/security you added.
6. A **3–5 min demo** (video or live) and updated role credentials.

> **Golden rule:** depth over breadth. One real integration + one solid multi-agent flow + honest evaluation beats five half-built features.